

WEBARENA: A REALISTIC WEB ENVIRONMENT FOR BUILDING AUTONOMOUS AGENTS

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ABSTRACT

With advances in generative AI, there is now potential for autonomous agents to manage daily tasks via natural language commands. However, current agents are primarily created and tested in simplified synthetic environments, leading to a disconnect with real-world scenarios. In this paper, we build an environment for language-guided agents that is *highly realistic* and *reproducible*. Specifically, we focus on agents that perform tasks on the web, and create an environment with fully functional websites from four common domains: e-commerce, social forum discussions, collaborative software development, and content management. Our environment is enriched with tools (*e.g.*, a map) and external knowledge bases (*e.g.*, user manuals) to encourage human-like task-solving. Building upon our environment, we release a set of benchmark tasks focusing on evaluating the *functional correctness* of task completions. The tasks in our benchmark are diverse, long-horizon, and designed to emulate tasks that humans routinely perform on the internet. We experiment with several baseline agents, integrating recent techniques such as reasoning before acting. The results demonstrate that solving complex tasks is challenging: our best GPT-4-based agent only achieves an end-to-end task success rate of 14.41%, significantly lower than the human performance of 78.24%. These results highlight the need for further development of robust agents, that current state-of-the-art large language models are far from perfect performance in these real-life tasks, and that WebArena can be used to measure such progress. Our code, data, environment reproduction resources, and video demonstrations are publicly available at <https://webarena.dev/>.

1 INTRODUCTION

Autonomous agents that perform everyday tasks via human natural language commands could significantly augment human capabilities, improve efficiency, and increase accessibility. Nonetheless, to fully leverage the power of autonomous agents, it is crucial to understand their behavior within an environment that is both *authentic* and *reproducible*. This will allow measurement of the ability of agents on tasks that human users care about in a fair and consistent manner.

Current environments for evaluate agents tend to *over-simplify* real-world situations. As a result, the functionality of many environments is a limited version of their real-world counterparts, leading to a lack of task diversity (Shi et al., 2017; Anderson et al., 2018; Gordon et al., 2018; Misra et al., 2016; Shridhar et al., 2020; 2021; Yao et al., 2022a). In addition, these simplifications often lower the complexity of tasks as compared to their execution in the real world (Puig et al., 2018; Shridhar et al., 2020; Yao et al., 2022a). Finally, some environments are presented as a static resource (Shi et al., 2017; Deng et al., 2023) where agents are confined to accessing only those states that were previously cached during data collection, thus limiting the breadth and diversity of exploration. For evaluation, many environments focus on comparing the textual *surface form* of the predicted

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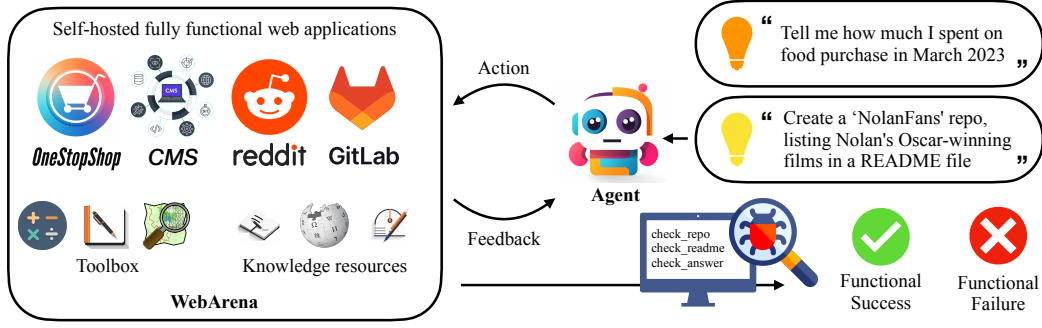


Figure 1: WebArena is a standalone, self-hostable web environment for building autonomous agents. WebArena creates websites from four popular categories with functionality and data mimicking their real-world equivalents. To emulate human problem-solving, WebArena also embeds tools and knowledge resources as independent websites. WebArena introduces a benchmark on interpreting *high-level realistic* natural language command to concrete web-based interactions. We provide annotated programs designed to programmatically validate the functional correctness of each task.

action sequences with reference action sequences, disregarding the *functional correctness* of the executions and possible alternative solutions (Puig et al., 2018; Jernite et al., 2019; Xu et al., 2021; Li et al., 2020; Deng et al., 2023). These limitations often result in a discrepancy between simulated environments and the real world, and can potentially impact the generalizability of AI agents to successfully understand, adapt, and operate within complex real-world situations.

We introduce WebArena, a *realistic* and *reproducible* web environment designed to facilitate the development of autonomous agents capable of executing tasks (§2). An overview of WebArena is in Figure 1. Our environment comprises four fully operational, self-hosted web applications, each representing a distinct domain prevalent on the internet: online shopping, discussion forums, collaborative development, and business content management. Furthermore, WebArena incorporates several utility tools, such as map, calculator, and scratchpad, to best support possible human-like task executions. Lastly, WebArena is complemented by an extensive collection of documentation and knowledge bases that vary from general resources like English Wikipedia to more domain-specific references, such as manuals for using the integrated development tool (Fan et al., 2022). The content populating these websites is extracted from their real-world counterparts, preserving the authenticity of the content served on each platform. We deliver the hosting services using Docker containers with gym-APIs (Brockman et al., 2016), ensuring both the usability and the reproducibility of WebArena.

Along with WebArena, we release a ready-to-use benchmark with 812 long-horizon web-based tasks (§3). Each task is described as a high-level natural language intent, emulating the abstract language usage patterns typically employed by humans (Bisk et al., 2019). Two example intents are shown in the upper left of Figure 1. We focus on evaluating the *functional correctness* of these tasks, *i.e.*, does the result of the execution actually achieve the desired goal (§3.2). For instance, to evaluate the example in Figure 2, our evaluation method verifies the concrete contents in the designated repository. This evaluation is not only more reliable (Zhong et al., 2017; Chen et al., 2021; Wang et al., 2022) than comparing the textual surface-form action sequences (Puig et al., 2018; Deng et al., 2023) but also accommodate a range of potential valid paths to achieve the same goal, which is a ubiquitous phenomenon in sufficiently complex tasks.

We use this benchmark to evaluate several agents that can follow NL command and perform web-based tasks (§4). These agents are implemented in a few-shot in-context learning fashion with powerful large language models (LLMs) such as GPT-4 and PALM-2. Experiment results show that the best GPT-4 agent performance is somewhat limited, with an end-to-end task success rate of only 14.41%, while the human performance is 78.24%. We hypothesize that the limited performance of current LLMs stems from a lack of crucial capabilities such as active exploration and failure recovery to successfully perform complex tasks (§5.2). These outcomes underscore the necessity for further development towards robust and effective agents (LeCun, 2022) in WebArena.

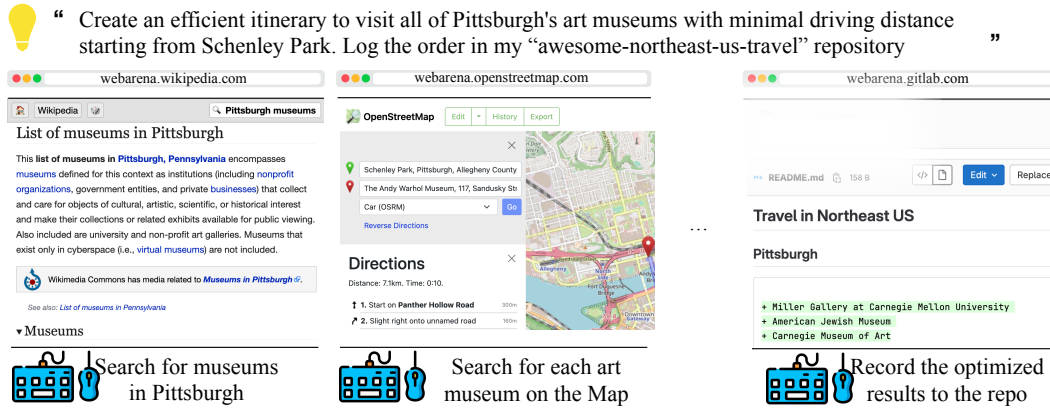


Figure 2: A high-level task that can be fully executed in WebArena. Success requires sophisticated, long-term planning and reasoning. To accomplish the goal (top), an agent needs to (1) find Pittsburgh art museums on Wikipedia, (2) identify their locations on a map (while optimizing the itinerary), and (3) update the README file in the appropriate repository with the planned route.

2 WEBARENA: WEBSITES AS AN ENVIRONMENT FOR AUTONOMOUS AGENTS

Our goal is to create a *realistic* and *reproducible* web environment. We achieve reproducibility by making the environment standalone, without relying on live websites. This circumvents technical challenges such as bots being subject to CAPTCHAs, unpredictable content modifications, and configuration changes, which obstruct a fair comparison across different systems over time. We achieve realism by using open-source libraries that underlie many in-use sites from several popular categories and importing data to our environment from their real-world counterparts.

2.1 CONTROLLING AGENTS THROUGH HIGH-LEVEL NATURAL LANGUAGE

The WebArena environment is denoted as $\mathcal{E} = \langle \mathcal{S}, \mathcal{A}, \mathcal{O}, \mathcal{T} \rangle$ with state space $\mathcal{S}$, action space $\mathcal{A}$ (§2.4) and observation space $\mathcal{O}$ (§2.3). The transition function $\mathcal{T} : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{S}$ is deterministic, and it is defined by the underlying implementation of each website in the environment. Performing a task described by a natural language intent i can be formulated as a partially observable Markov decision process (POMDP): at each time step t , an agent issues an action $a_t \in \mathcal{A}$ given the partial observation $o_t \in \mathcal{O}$. Consequently, the action results in a new state $s_{t+1} \in \mathcal{S}$ and its corresponding observation $o_{t+1} \in \mathcal{O}$. We propose a reward function $r(a, s)$ to measure the success of a task execution, where a represents the sequence of actions, and s denotes all intermediate states. This reward function assesses if state transitions align with the expectations of the intents. For example, with an intent to place an order, it verifies whether an order has been placed. Additionally, it evaluates the accuracy of the agent’s actions, such as checking the correctness of the predicted answer.

2.2 WEBSITE SELECTION

To decide which categories of websites to use, we first analyzed approximately 200 examples from the authors’ actual web browser histories. Each author delved into their browsing histories, summarizing the goal of particular segments of their browser session. Based on this, we classified the visited websites into abstract categories. We then identified the four most salient categories and implemented one instance per category based on this analysis: (1) E-commerce platforms supporting online shopping activities (e.g., Amazon, eBay), (2) social forum platforms for opinion exchanges (e.g., Reddit, StackExchange), (3) collaborative development platforms for software development (e.g., GitLab), and (4) content management systems (CMS) that manage the creation and revision of the digital content (e.g., online store management).

In addition to these platforms, we selected three utility-style tools that are frequently used in web-based tasks: (1) a map for navigation and searching for information about points of interest (POIs) such as institutions or locations (2) a calculator, and (3) a scratchpad for taking notes. As information-seeking and knowledge acquisition are critical in web-based tasks, we also incorporated various knowledge resources into WebArena. These resources range from general information hubs, such as the English Wikipedia, to more specialized knowledge bases, such as the website user manuals.

²https://developer.mozilla.org/en-US/docs/Glossary/Accessibility_tree

[1582] clicks the button given the observation of [1582] Add to Cart. This flexible element selection allows WebArena to support agents designed in various ways (e.g., accepting input from different modalities) without compromising fair comparison metrics such as step count.

User Role Simulation Users of the same website often have disparate experiences due to their distinct *roles*, *permissions*, and *interaction histories*. We emulate this scenario by generating unique user profiles on each platform. The details can be found in Appendix A.3.

3 BENCHMARK SUITE OF WEB-BASED TASKS

We provide a benchmark with 812 test examples on grounding high-level natural language instructions to interactions in WebArena. Each example has a metric to evaluate the functional correctness of the task execution. In this section, we first formally define the task of controlling an autonomous agent through natural language. Then we introduce the annotation process of our benchmark.

3.1 INTENT COLLECTION

We focus on curating *realistic* intents to carry out *complex* and *creative* tasks within WebArena. To start with, our annotators were guided to spend a few minutes exploring the websites to familiarize themselves with the websites’ content and functionalities. As most of our websites are virtually identical to their open-web counterparts, despite having sampled data, most annotators can quickly comprehend the websites.

Next, we instructed the annotators to formulate intents based on the following criteria:

- (1) The intent should be *abstract* and *high-level*, implying that the task cannot be fulfilled with merely one or two actions. As an example, instead of “click the *science* subreddit”, we encouraged annotators to come up with something more complex like “post a greeting message on *science* subreddit”, which involves performing multiple actions.
- (2) The intent should be *creative*. Common tasks such as account creation can be easily thought of. We encouraged the annotators to add constraints (e.g., “create a Reddit account *identical to my GitLab one*”) to make the intents more unique.
- (3) The intent should be formulated as a *template* by making replaceable elements as variables. The annotators were also responsible for developing several instantiations for each variable. For example, the intent “create a Reddit account identical to my GitLab one” can be converted into “create a {{site1}} account identical to my {{site2}} one”, with an instantiation like “{site1: Reddit, site2: GitLab}” and another like “{site1: GitLab, site2: OneStopShopping}”. Notably, tasks derived from the same template can have distinct execution traces. The similarity resides primarily in the high-level semantics rather than the specific implementation.

We also provided a prompt for the annotators to use with ChatGPT³ for inspiration, that contains an overview of each website and instructs the model to describe potential tasks to be performed on these sites. Furthermore, we offered a curated list of examples for annotators to reference.

Intent Analysis In total, we curated 241 templates and 812 instantiated intents. On average, each template is instantiated to 3.3 examples. The intent distribution is shown in Figure 6.

Furthermore, we classify the intents into three primary categories with examples shown in Figure 5:

- (1) **Information-seeking** tasks expect a textual response. Importantly, these tasks in WebArena often require navigation across multiple pages or focus on *user-centric* content. This makes them distinct from open-domain question-answering (Yang et al., 2018; Kwiatkowski et al., 2019), which focuses on querying general knowledge with a simple retrieval step. For instance, to answer “When was the last time I bought the shampoo”, an agent traverses the user’s purchase history, checking order details to identify the most recent shampoo purchase.
- (2) **Site navigation**: This category is composed of tasks that require navigating through web pages using a variety of interactive elements such as search functions and links. The objective is often to locate specific information or navigate to a particular section of a site.

³<https://chat.openai.com/>

Action Type	Description
noop	Do nothing
click(elem)	Click at an element
hover(elem)	Hover on an element
type(elem, text)	Type to an element
press(key_comb)	Press a key comb
tab_focus(index)	focus on i -th tab
new_tab	Open a new tab
tab_close	Close current tab
go_back	Visit the last URL
go_forward	Undo go_back
goto(URL)	Go to URL

Figure 4: Action Space of WebArena

Category	Example
Information Seeking	When was the last time I bought shampoo
	Compare walking and driving time from AMC Waterfront to Randyland
Site Navigation	Checkout merge requests assigned to me
	Show me the ergonomic chair with the best rating
Content & Config	Post to ask “whether I need a car in NYC”
	Delete the reviews from the scammer Yoke

Figure 5: Example intents from three categories.

- (3) **Content and configuration operation:** This category encapsulates tasks that require operating in the web environment to create, revise, or configure content or settings. This includes adjusting settings, managing accounts, performing online transactions, generating new web content, and modifying existing content. Examples range from updating a social media status or README file to conducting online purchases and configuring privacy settings.

3.2 EVALUATION ANNOTATION

Evaluating Information Seeking Tasks To measure the correctness of information-seeking tasks where a textual answer is expected, we provide the annotated answer a^* for each intent. The a^* is further compared with the predicted answer $\hat{a}$ with one of the following scoring functions $r_{\text{info}}(\hat{a}, a^*)$.

First, we define `exact_match` where only $\hat{a}$ that is identical with a^* receives a score of one. This function is primarily applicable to intent types whose responses follow a more standardized format, similar to the evaluation on question answering literature (Rajpurkar et al., 2016; Yang et al., 2018).

Second, we create `must_include` where any $\hat{a}$ containing a^* receives a score of one. This function is primarily used in when an unordered list of text is expected or where the emphasis of evaluation is on certain key concepts. In the second example in Table 1, we expect both the correct name and the email address to be presented, irrespective of the precise wording used to convey the answer.

Finally, we introduce `fuzzy_match` where we utilize a language model to assess whether $\hat{a}$ is semantically equivalent to a^* . Specifically, in this work, we use `gpt-4-0613` to perform this evaluation. The corresponding prompt details are provided in Appendix A.6. The `fuzzy_match` function applies to situations where the format of the answer is diverse. For instance, in responding to “*Compare the time for walking and driving route from AMC Waterfront to Randyland*”, it is essential to ensure that driving time and walking time are accurately linked with the correct terms. The `fuzzy_match` function could also flexibly match the time “2h58min” with different forms such as “2 hour 58 minutes”, “2:58” and others.

Evaluating Site Navigation and Content & Config Tasks The tasks in these categories require accessing web pages that meet certain conditions or performing operations that modify the underlying data storage of the respective websites. To assess these, we establish reward functions $r_{\text{prog}}(\mathbf{s})$ that programmatically examine the intermediate states $\mathbf{s}$ within an execution trajectory to ascertain whether the outcome aligns with the intended result. These intermediate states are often the underlying databases of the websites, the status, and the content of a web page at each step of the execution.

Evaluating each instance involves two components. First, we provide a `locator`, tasked with retrieving the critical content pertinent to each intent. The implementation of this locator varies from a database query, a website-supported API call, to a JavaScript element selection on the relevant web page, depending on implementation feasibility. For example, the evaluation process for the intent of the fifth example in Table 1, first obtains the URL of the latest post by examining the last state in the state sequence $\mathbf{s}$. Then it navigates to the corresponding post page and obtains the post’s content by running the Javascript “`document.querySelector('.submission__inner').outerText`”.

Subsequently, we annotate `keywords` that need to exist within the located content. For example, the evaluation verifies if the post is correctly posted in the “nyc” subreddit by examining the URL of

Function	ID	Intent	Eval Implementation
$r_{\text{info}}(a^*, \hat{a})$	1	Tell me the name of the customer who has the most cancellations in the history	<code>exact_match($\hat{a}$, "Samantha Jones")</code>
	2	Find the customer name and email with phone number 8015551212	<code>must_include($\hat{a}$, "Sean Miller")</code> <code>must_include($\hat{a}$, "sean@gmail.com")</code>
	3	Compare walking and driving time from AMC Waterfront to Randyland	<code>fuzzy_match($\hat{a}$, "walking: 2h58min")</code> <code>fuzzy_match($\hat{a}$, "driving: 21min")</code>
$r_{\text{prog}}(s)$	4	Checkout merge requests assigned to me	<code>url=locate_current_url(s)</code> <code>exact_match(URL, "gitlab.com/merge_requests?assignee_username=byteblaze")</code>
	5	Post to ask "whether I need a car in NYC"	<code>url=locate_latest_post_url(s)</code> <code>body=locate_latest_post_body(s)</code> <code>must_include(URL, "/f/nyc")</code> <code>must_include(body, "a car in NYC")</code>

Table 1: We introduce two evaluation approaches. r_{info} (top) measures the correctness of performing information-seeking tasks. It compares the predicted answer $\hat{a}$ with the annotated reference a^* with three implementations. r_{prog} (bottom) programmatically checks whether the intermediate states during the executions possess the anticipated properties specified by the intent.

the post and if the post contains the requested content by examining the post content. We reuse the `exact_match` and `must_include` functions from information-seeking tasks for this purpose.

Unachievable Tasks Due to constraints such as inadequate evidence, user permissions (§A.3), or the absence of necessary functional support on the website, humans may ask for tasks that are not possible to complete. Inspired by previous work on evaluating question-answering models on unanswerable questions (Rajpurkar et al., 2018), we design unachievable tasks in WebArena. For instance, fulfilling an intent like “*Tell me the contact number of OneStopShop*” is impracticable in WebArena, given that the website does not provide such contact information. We label such instances as "N/A" and expect an agent to produce an equivalent response. These examples allow us to assess an agent’s ability to avoid making unfounded claims and its adherence to factual accuracy.

Annotation Process The intents were contributed by the authors following the annotation guideline in §3.1. Every author has extensive experience with web-based tasks. The reference answers to the information-seeking tasks were curated by the authors and an external annotator. To ensure consistency and accuracy, each question was annotated twice. If the two annotators disagreed, a third annotator finalized the annotation. The programs to evaluate the remaining examples were contributed by three of the authors who are proficient in JavaScript programming. Difficult tasks were often discussed collectively to ensure the correctness of the annotation. The annotation required the annotator to undertake the full execution and scrutinize the intermediate states.

Human Performance We sample one task from each of the 170 templates and ask five computer science graduate students to perform these tasks. The human performance is on the right. Overall, the human annotators complete 78.24% of the tasks, with lower performance on information-seeking tasks. Through examining the recorded trajectories, we found that 50% of the failures are due to misinterpreting the intent (*e.g.*, providing travel distance when asked for travel time), incomplete answers (*e.g.*, providing only name when asked for name and email), and incomplete executions (*e.g.*, partially filling the product information), while the remaining instances have more severe failures, where the executions are off-target.

Avg. Time	110s
Success Rate _{info}	74.68%
Success Rate _{others}	81.32%
Success Rate _{all}	78.24%

4 BASELINE WEB AGENTS

We experiment with three LLMs using two prompting strategies, both with two examples in the context. In the first setting, we ask the LLM to directly predict the next action given the current observation, the intent and the previously performed action. In the second setting, with the same information, the model first performs chain-of-thought reasoning steps in the text before the action prediction (CoT, Wei et al. (2022); Yao et al. (2022b)). Before the examples, we provide a detailed

overview of the browser environment, the allowed actions, and many rules. To make the model aware of the unachievable tasks, the instruction explicitly asks the agent to stop if it believes the task is impossible to perform. We refer to this directive as Unachievable hint, or **UA hint**. This introduction is largely identical to the guidelines we presented to human annotators to ensure a fair comparison. We use an accessibility tree with element IDs as the observation space. The agent can identify which element to interact with by the ID of the element. For instance, the agent can issue `click [1582]` to click the “Add to Cart” button with the ID of 1582. The full prompts can be found in Appendix A.7. The detailed configurations of each model can be found in Appendix A.5.

5 RESULTS

5.1 MAIN RESULTS

The main results are shown on the top of Table 2. GPT-4 (OpenAI, 2023) with CoT prompting achieves a modest end-to-end task success rate of 11.70%, which is significantly lower than the human performance of 78.24%. GPT-3.5 (OpenAI, 2022) with CoT prompting is only able to successfully perform 8.75% of the tasks. The explicit reasoning procedure is somewhat helpful, it brings 2.34% improvement over the version without it. Further, TEXT-BISON-001 (Anil et al., 2023) underperforms GPT-3.5, with a success rate of 5.05%. These results underline the inherent challenges and complexities of executing tasks that span long horizons, particularly in realistic environments such as WebArena.

CoT	UA Hint	Model	SR	SR _{AC}	SR _{UA}
✓	✓	TEXT-BISON-001	5.05	4.00	27.78
✗	✓	GPT-3.5	6.41	4.90	38.89
✓	✓	GPT-3.5	8.75	6.44	58.33
✓	✓	GPT-4	11.70	8.63	77.78
✗	✗	GPT-3.5	5.10	4.90	8.33
✓	✗	GPT-3.5	6.16	6.06	8.33
✓	✗	GPT-4	14.41	13.02	44.44
-	✓	Human	78.24	77.30	100.00

Table 2: The end-to-end task success rate (SR %) on WebArena with different prompting strategies. **CoT**: the model performs step-by-step reasoning before issuing the action. **UA hint**: ask the model to stop when encountering unachievable questions.

5.2 ANALYSIS

Do models know when to stop? In our error analysis of the execution trajectories, we observe a prevalent error pattern of early stopping due to the model’s conclusion of unachievability. For instance, GPT-4 erroneously identifies 54.9% of feasible tasks as impossible. This issue primarily stems from the UA hint in the instruction, while this hint allows models to identify unachievable tasks, it also hinders performance on achievable tasks. To address this, we conduct an ablation study where we remove this hint. We then break down the success rate for both achievable and unachievable tasks. As shown in Table 2, eliminating this instruction led to a performance boost in achievable tasks, enhancing the overall task success rate of GPT-4 to 14.41%. Despite an overall decline in identifying unachievable tasks, GPT-4 retains the capacity to recognize 44.44% of such tasks. It does so by generating *reasons of non-achievability*, even without explicit instructions. On the other hand, GPT-3.5 rarely exhibits this level of reasoning. Instead, it tends to follow problematic patterns such as hallucinating incorrect answers, repeating invalid actions, or exceeding the step limits. This result suggests that even subtle differences in instruction design can significantly influence the behavior of a model in performing interactive tasks in complex environments.

Can a model maintain consistent performance across similar tasks? Tasks that originate from the same template usually follow similar reasoning and planning processes, even though their observations and executions will differ. We plot a histogram of per-template success rates for our models in Table 3. Of the 61 templates, GPT-4 manages to achieve a 100% task success rate on only four templates, while GPT-3.5 fails to achieve full task completion for any of the templates. In many cases, the models are only able to complete one task variation with a template. These observations indicate that even when tasks are derived from the same template, they can present

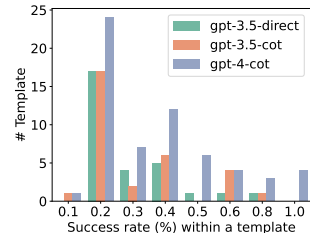


Table 3: Distribution of success rates on templates with ≥ 1 successful executions on GPT models (no UA hint).

Benchmark	Dynamic Interaction?	Realistic Environment?	Diverse Human Tasks?	Functional Correctness?
Mind2Web (Deng et al., 2023)	✗	✓	✓	✗
Form/QAWoB (Shi et al., 2017)	✗	✓	✓	✗
MiniWoB++ (Liu et al., 2018)	✓	✗	✗	✓
Webshop (Yao et al., 2022a)	✓	✗	✗	✓
ALFRED (Shridhar et al., 2020)	✓	✗	✗	✓
VirtualHome (Puig et al., 2018)	✗	✗	✓	✗
AndroidEnv (Toyama et al., 2021)	✓	✓	✗	✗
WebArena	✓	✓	✓	✓

Table 4: The comparison between our benchmark and existing benchmarks on grounding natural language instructions to concrete executions. Our benchmark is implemented in our fully interactable highly-realistic environment. It features diverse tasks humans may encounter in their daily routines. We design evaluation metrics to assess the functional correctness of task executions.

distinct challenges. For instance, while “*Fork metaseq*” can be a straightforward task, “*Fork all repos from Facebook*” derived from the same template requires more repetitive operations, hence increasing its complexity. Therefore, WebArena provide a testbed to evaluate more sophisticated methods. In particular, those that incorporate memory components, enabling the *reuse* of successful strategies from past experiments Zhou et al. (2022a); Wang et al. (2023). More error analysis with examples can be found in Appendix A.8.

6 RELATED WORK

Benchmarks for Controlling Agents through Natural Language Controlling agents via natural language in the digital world have been studied in the literature (Branavan et al., 2009; Shi et al., 2017; Liu et al., 2018; Toyama et al., 2021; Deng et al., 2023; Li et al., 2020; Xu et al., 2021). However, the balance between *functionality*, *authenticity*, and *support for environmental dynamics* remains a challenge. Existing benchmarks often compromise these aspects, as shown in Table 4. Some works rely on static states, limiting agents’ explorations and functional correctness evaluation (Shi et al., 2017; Deng et al., 2023), while others simplify real-world complexities, restricting task variety (Yao et al., 2022a; Liu et al., 2018). While AndroidEnv (Toyama et al., 2021) replicates an Android setup, it does not evaluate complex and realistic tasks. This gap is also seen in synthetic environments (Kolve et al., 2017; Shridhar et al., 2020; Puig et al., 2018) and extends to gaming environments (Fan et al., 2022; Küttler et al., 2020), where the environment mechanisms often diverge from human objectives.

Interactive Decision-Making Agents Nakano et al. (2021) introduce WebGPT which searches the web and reads the search results to answer questions. Gur et al. (2023) propose a web agent that decomposes tasks into more manageable sub-tasks and synthesizes Javascript code for the task executions. Adding a multi-modal dimension, Lee et al. (2023) and Shaw et al. (2023) develop agents that predict actions based on screenshots of web pages rather than relying on the text-based DOM trees. Performing tasks in interactive environments requires the agents to exhibit several capabilities including hierarchical planning, state tracking, and error recovery. Existing works (Huang et al., 2022; Madaan et al., 2022; Li et al., 2023) observe LLMs could break a task into more manageable sub-tasks (Zhou et al., 2022b). This process can be further refined by representing task executions as programs, a technique that aids sub-task management and skill reuse (Zhou et al., 2022a; Liang et al., 2023; Wang et al., 2023; Gao et al., 2023). Meanwhile, search and backtracking methods introduce a more structured approach to planning while also allowing for decision reconsideration (Yao et al., 2023; Long, 2023). Existing works also incorporate failure recovery, self-correction (Shinn et al., 2023; Kim et al., 2023), observation summarization (Sridhar et al., 2023) to improve execution robustness. The complexity of WebArena presents a unique challenge and opportunity for further testing and improvement of these methods.

7 CONCLUSION

We present *WebArena*, a highly-realistic, standalone, and reproducible web environment designed for the development and testing of autonomous agents. *WebArena* includes fully functional web applications and genuine data from four major categories, providing a realistic platform for agent interaction. It further supports a wide range of tools and external knowledge bases, fostering a focus on human-like problem-solving. Additionally, we curate a comprehensive benchmark consisting of 812 examples that focus on translating high-level natural language intents into specific web interactions. We also offer metrics to programmatically ascertain whether tasks have been completed according to the desired objectives. Our experiments show that even GPT-4 only achieves a limited end-to-end task success rate of 14.41%, significantly lagging behind the human performance of 78.24%. These findings underscore the need for future research to focus on enhancing the robustness and efficacy of autonomous agents within *WebArena* environment.

ACKNOWLEDGEMENT

We would like to thank Emmy Liu, Zhiruo Wang, Zhitong Guo for examining our annotations, Shunyu Yao for providing the raw Amazon product data in Webshop, Pengfei Liu, Zaid Sheikh and Aman Madaan for the helpful discussions. We are also grateful to the Center for AI Safety for providing computational resources. This material is partly based on research sponsored in part by the Air Force Research Laboratory under agreement number FA8750-19-2-0200. The U.S. Government is authorized to reproduce and distribute reprints for Governmental purposes notwithstanding any copyright notation thereon. The views and conclusions contained herein are those of the authors and should not be interpreted as necessarily representing the official policies or endorsements, either expressed or implied, of the Air Force Research Laboratory or the U.S. Government. This project was also partially supported by a gift from AWS AI.

REFERENCES

- Peter Anderson, Qi Wu, Damien Teney, Jake Bruce, Mark Johnson, Niko Sünderhauf, Ian D. Reid, Stephen Gould, and Anton van den Hengel. Vision-and-language navigation: Interpreting visually-grounded navigation instructions in real environments. In *2018 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2018, Salt Lake City, UT, USA, June 18-22, 2018*, pp. 3674–3683. IEEE Computer Society, 2018. doi: 10.1109/CVPR.2018.00387. URL http://openaccess.thecvf.com/content_cvpr_2018/html/Anderson_Vision-and-Language_Navigation_Interpreting_CVPR_2018_paper.html.
- Rohan Anil, Andrew M. Dai, Orhan Firat, Melvin Johnson, Dmitry Lepikhin, Alexandre Passos, Siamak Shakeri, Emanuel Taropa, Paige Bailey, Zhifeng Chen, Eric Chu, Jonathan H. Clark, Laurent El Shafey, Yanping Huang, Kathy Meier-Hellstern, Gaurav Mishra, Erica Moreira, Mark Omernick, Kevin Robinson, Sebastian Ruder, Yi Tay, Kefan Xiao, Yuanzhong Xu, Yujing Zhang, Gustavo Hernandez Abrego, Junwhan Ahn, Jacob Austin, Paul Barham, Jan Botha, James Bradbury, Siddhartha Brahma, Kevin Brooks, Michele Catasta, Yong Cheng, Colin Cherry, Christopher A. Choquette-Choo, Aakanksha Chowdhery, Clément Crepy, Shachi Dave, Mostafa Dehghani, Sunipa Dev, Jacob Devlin, Mark Díaz, Nan Du, Ethan Dyer, Vlad Feinberg, Fangxiaoyu Feng, Vlad Fienber, Markus Freitag, Xavier Garcia, Sebastian Gehrmann, Lucas Gonzalez, Guy Gur-Ari, Steven Hand, Hadi Hashemi, Le Hou, Joshua Howland, Andrea Hu, Jeffrey Hui, Jeremy Hurwitz, Michael Isard, Abe Ittycheriah, Matthew Jagielski, Wenhao Jia, Kathleen Kenealy, Maxim Krikun, Sneha Kudugunta, Chang Lan, Katherine Lee, Benjamin Lee, Eric Li, Music Li, Wei Li, YaGuang Li, Jian Li, Hyeontaek Lim, Hanzhao Lin, Zhongtao Liu, Frederick Liu, Marcello Maggioni, Aroma Mahendru, Joshua Maynez, Vedant Misra, Maysam Moussalem, Zachary Nado, John Nham, Eric Ni, Andrew Nystrom, Alicia Parrish, Marie Pellat, Martin Polacek, Alex Polozov, Reiner Pope, Siyuan Qiao, Emily Reif, Bryan Richter, Parker Riley, Alex Castro Ros, Aurko Roy, Brennan Saeta, Rajkumar Samuel, Renee Shelby, Ambrose Slone, Daniel Smilkov, David R. So, Daniel Sohn, Simon Tokumine, Dasha Valter, Vijay Vasudevan, Kiran Vodrahalli, Xuezhi Wang, Pidong Wang, Zirui Wang, Tao Wang, John Wieting, Yuhuai Wu, Kelvin Xu, Yunhan Xu, Linting Xue, Pengcheng Yin, Jiahui Yu, Qiao Zhang, Steven Zheng, Ce Zheng, Weikang Zhou, Denny Zhou, Slav Petrov, and Yonghui Wu. Palm 2 technical report, 2023.

- Yonatan Bisk, Jan Buys, Karl Pichotta, and Yejin Choi. Benchmarking hierarchical script knowledge. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pp. 4077–4085, Minneapolis, Minnesota, 2019. Association for Computational Linguistics. doi: 10.18653/v1/N19-1412. URL <https://aclanthology.org/N19-1412>.
- S.R.K. Branavan, Harr Chen, Luke Zettlemoyer, and Regina Barzilay. Reinforcement learning for mapping instructions to actions. In *Proceedings of the Joint Conference of the 47th Annual Meeting of the ACL and the 4th International Joint Conference on Natural Language Processing of the AFNLP*, pp. 82–90, Suntec, Singapore, 2009. Association for Computational Linguistics. URL <https://aclanthology.org/P09-1010>.
- Greg Brockman, Vicki Cheung, Ludwig Pettersson, Jonas Schneider, John Schulman, Jie Tang, and Wojciech Zaremba. Openai gym, 2016. URL <https://arxiv.org/abs/1606.01540>.
- Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde, Jared Kaplan, Harri Edwards, Yura Burda, Nicholas Joseph, Greg Brockman, et al. Evaluating large language models trained on code. *ArXiv preprint*, abs/2107.03374, 2021. URL <https://arxiv.org/abs/2107.03374>.
- Xiang Deng, Yu Gu, Boyuan Zheng, Shijie Chen, Samuel Stevens, Boshi Wang, Huan Sun, and Yu Su. Mind2web: Towards a generalist agent for the web, 2023.
- Linxi Fan, Guanzhi Wang, Yunfan Jiang, Ajay Mandlekar, Yuncong Yang, Haoyi Zhu, Andrew Tang, De-An Huang, Yuke Zhu, and Anima Anandkumar. Minedojo: Building open-ended embodied agents with internet-scale knowledge. In *Thirty-sixth Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2022. URL https://openreview.net/forum?id=rc8o_j8I8PX.
- Luyu Gao, Aman Madaan, Shuyan Zhou, Uri Alon, Pengfei Liu, Yiming Yang, Jamie Callan, and Graham Neubig. Pal: Program-aided language models. In *International Conference on Machine Learning*, pp. 10764–10799. PMLR, 2023.
- Daniel Gordon, Aniruddha Kembhavi, Mohammad Rastegari, Joseph Redmon, Dieter Fox, and Ali Farhadi. IQA: visual question answering in interactive environments. In *2018 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2018, Salt Lake City, UT, USA, June 18-22, 2018*, pp. 4089–4098. IEEE Computer Society, 2018. doi: 10.1109/CVPR.2018.00430. URL http://openaccess.thecvf.com/content_cvpr_2018/html/Gordon_IQA_Visual_Question_CVPR_2018_paper.html.
- Izzeddin Gur, Hiroki Furuta, Austin Huang, Mustafa Safdari, Yutaka Matsuo, Douglas Eck, and Aleksandra Faust. A real-world webagent with planning, long context understanding, and program synthesis. *arXiv preprint arXiv:2307.12856*, 2023.
- Wenlong Huang, Pieter Abbeel, Deepak Pathak, and Igor Mordatch. Language models as zero-shot planners: Extracting actionable knowledge for embodied agents. In Kamalika Chaudhuri, Stefanie Jegelka, Le Song, Csaba Szepesvári, Gang Niu, and Sivan Sabato (eds.), *International Conference on Machine Learning, ICML 2022, 17-23 July 2022, Baltimore, Maryland, USA*, volume 162 of *Proceedings of Machine Learning Research*, pp. 9118–9147. PMLR, 2022. URL <https://proceedings.mlr.press/v162/huang22a.html>.
- Yacine Jernite, Kavya Srinet, Jonathan Gray, and Arthur Szlam. CraftAssist Instruction Parsing: Semantic Parsing for a Minecraft Assistant. *ArXiv preprint*, abs/1905.01978, 2019. URL <https://arxiv.org/abs/1905.01978>.
- Geunwoo Kim, Pierre Baldi, and Stephen McAleer. Language models can solve computer tasks. *ArXiv preprint*, abs/2303.17491, 2023. URL <https://arxiv.org/abs/2303.17491>.
- Eric Kolve, Roozbeh Mottaghi, Winson Han, Eli VanderBilt, Luca Weihs, Alvaro Herrasti, Daniel Gordon, Yuke Zhu, Abhinav Gupta, and Ali Farhadi. AI2-THOR: An Interactive 3D Environment for Visual AI. *arXiv*, 2017.

- Heinrich Küttler, Nantas Nardelli, Alexander H. Miller, Roberta Raileanu, Marco Selvatici, Edward Grefenstette, and Tim Rocktäschel. The nethack learning environment. In Hugo Larochelle, Marc’Aurelio Ranzato, Raia Hadsell, Maria-Florina Balcan, and Hsuan-Tien Lin (eds.), *Advances in Neural Information Processing Systems 33: Annual Conference on Neural Information Processing Systems 2020, NeurIPS 2020, December 6-12, 2020, virtual*, 2020. URL <https://proceedings.neurips.cc/paper/2020/hash/569ff987c643b4bedf504efda8f786c2-Abstract.html>.
- Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Jacob Devlin, Kenton Lee, Kristina Toutanova, Llion Jones, Matthew Kelcey, Ming-Wei Chang, Andrew M. Dai, Jakob Uszkoreit, Quoc Le, and Slav Petrov. Natural questions: A benchmark for question answering research. *Transactions of the Association for Computational Linguistics*, 7:452–466, 2019. doi: 10.1162/tacl_a_00276. URL <https://aclanthology.org/Q19-1026>.
- Yann LeCun. A path towards autonomous machine intelligence version 0.9. 2, 2022-06-27. *Open Review*, 62, 2022.
- Kenton Lee, Mandar Joshi, Iulia Raluca Turc, Hexiang Hu, Fangyu Liu, Julian Martin Eisenschlos, Urvashi Khandelwal, Peter Shaw, Ming-Wei Chang, and Kristina Toutanova. Pix2struct: Screenshot parsing as pretraining for visual language understanding. In *International Conference on Machine Learning*, pp. 18893–18912. PMLR, 2023.
- Xinze Li, Yixin Cao, Muhao Chen, and Aixin Sun. Take a break in the middle: Investigating subgoals towards hierarchical script generation. *ArXiv preprint*, abs/2305.10907, 2023. URL <https://arxiv.org/abs/2305.10907>.
- Yang Li, Jiacong He, Xin Zhou, Yuan Zhang, and Jason Baldridge. Mapping natural language instructions to mobile UI action sequences. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 8198–8210, Online, 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.acl-main.729. URL <https://aclanthology.org/2020.acl-main.729>.
- Jacky Liang, Wenlong Huang, Fei Xia, Peng Xu, Karol Hausman, Brian Ichter, Pete Florence, and Andy Zeng. Code as policies: Language model programs for embodied control. In *2023 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 9493–9500. IEEE, 2023.
- Evan Zheran Liu, Kelvin Guu, Panupong Pasupat, Tianlin Shi, and Percy Liang. Reinforcement learning on web interfaces using workflow-guided exploration. In *6th International Conference on Learning Representations, ICLR 2018, Vancouver, BC, Canada, April 30 - May 3, 2018, Conference Track Proceedings*. OpenReview.net, 2018. URL <https://openreview.net/forum?id=ryTp3f-0->.
- Jieyi Long. Large language model guided tree-of-thought. *ArXiv preprint*, abs/2305.08291, 2023. URL <https://arxiv.org/abs/2305.08291>.
- Aman Madaan, Shuyan Zhou, Uri Alon, Yiming Yang, and Graham Neubig. Language models of code are few-shot commonsense learners. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pp. 1384–1403, Abu Dhabi, United Arab Emirates, 2022. Association for Computational Linguistics. URL <https://aclanthology.org/2022.emnlp-main.90>.
- Dipendra K Misra, Jaeyong Sung, Kevin Lee, and Ashutosh Saxena. Tell me dave: Context-sensitive grounding of natural language to manipulation instructions. *The International Journal of Robotics Research*, 35(1-3):281–300, 2016.
- Reiichiro Nakano, Jacob Hilton, Suchir Balaji, Jeff Wu, Long Ouyang, Christina Kim, Christopher Hesse, Shantanu Jain, Vineet Kosaraju, William Saunders, et al. Webgpt: Browser-assisted question-answering with human feedback. *arXiv preprint arXiv:2112.09332*, 2021.
- OpenAI. Chatgpt: Optimizing language models for dialogue. 2022.
- OpenAI. Gpt-4 technical report. *arXiv*, pp. 2303–08774, 2023.

- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*, 35: 27730–27744, 2022.
- Xavier Puig, Kevin Ra, Marko Boben, Jiaman Li, Tingwu Wang, Sanja Fidler, and Antonio Torralba. Virtualhome: Simulating household activities via programs. In *2018 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2018, Salt Lake City, UT, USA, June 18-22, 2018*, pp. 8494–8502. IEEE Computer Society, 2018. doi: 10.1109/CVPR.2018.00886. URL http://openaccess.thecvf.com/content_cvpr_2018/html/Puig_VirtualHome_Simulating_Household_CVPR_2018_paper.html.
- Pranav Rajpurkar, Jian Zhang, Konstantin Lopyrev, and Percy Liang. SQuAD: 100,000+ questions for machine comprehension of text. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, pp. 2383–2392, Austin, Texas, 2016. Association for Computational Linguistics. doi: 10.18653/v1/D16-1264. URL <https://aclanthology.org/D16-1264>.
- Pranav Rajpurkar, Robin Jia, and Percy Liang. Know what you don’t know: Unanswerable questions for SQuAD. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pp. 784–789, Melbourne, Australia, 2018. Association for Computational Linguistics. doi: 10.18653/v1/P18-2124. URL <https://aclanthology.org/P18-2124>.
- Peter Shaw, Mandar Joshi, James Cohan, Jonathan Berant, Panupong Pasupat, Hexiang Hu, Urvashi Khandelwal, Kenton Lee, and Kristina Toutanova. From pixels to ui actions: Learning to follow instructions via graphical user interfaces. *arXiv preprint arXiv:2306.00245*, 2023.
- Tianlin Shi, Andrej Karpathy, Linxi Fan, Jonathan Hernandez, and Percy Liang. World of bits: An open-domain platform for web-based agents. In Doina Precup and Yee Whye Teh (eds.), *Proceedings of the 34th International Conference on Machine Learning, ICML 2017, Sydney, NSW, Australia, 6-11 August 2017*, volume 70 of *Proceedings of Machine Learning Research*, pp. 3135–3144. PMLR, 2017. URL <http://proceedings.mlr.press/v70/shi17a.html>.
- Noah Shinn, Beck Labash, and Ashwin Gopinath. Reflexion: an autonomous agent with dynamic memory and self-reflection. *ArXiv preprint*, abs/2303.11366, 2023. URL <https://arxiv.org/abs/2303.11366>.
- Mohit Shridhar, Jesse Thomason, Daniel Gordon, Yonatan Bisk, Winson Han, Roozbeh Mottaghi, Luke Zettlemoyer, and Dieter Fox. ALFRED: A benchmark for interpreting grounded instructions for everyday tasks. In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2020, Seattle, WA, USA, June 13-19, 2020*, pp. 10737–10746. IEEE, 2020. doi: 10.1109/CVPR42600.2020.01075. URL <https://doi.org/10.1109/CVPR42600.2020.01075>.
- Mohit Shridhar, Xingdi Yuan, Marc-Alexandre Côté, Yonatan Bisk, Adam Trischler, and Matthew J. Hausknecht. Alfworld: Aligning text and embodied environments for interactive learning. In *9th International Conference on Learning Representations, ICLR 2021, Virtual Event, Austria, May 3-7, 2021*. OpenReview.net, 2021. URL <https://openreview.net/forum?id=0IOX0YcCdTn>.
- Abishek Sridhar, Robert Lo, Frank F Xu, Hao Zhu, and Shuyan Zhou. Hierarchical prompting assists large language model on web navigation. *arXiv preprint arXiv:2305.14257*, 2023.
- Daniel Toyama, Philippe Hamel, Anita Gergely, Gheorghe Comanici, Amelia Glaese, Zafarali Ahmed, Tyler Jackson, Shibl Mourad, and Doina Precup. Androidenv: A reinforcement learning platform for android. *ArXiv preprint*, abs/2105.13231, 2021. URL <https://arxiv.org/abs/2105.13231>.
- Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. Voyager: An open-ended embodied agent with large language models. *ArXiv preprint*, abs/2305.16291, 2023. URL <https://arxiv.org/abs/2305.16291>.

- Zhiruo Wang, Shuyan Zhou, Daniel Fried, and Graham Neubig. Execution-based evaluation for open-domain code generation. *ArXiv preprint*, abs/2212.10481, 2022. URL <https://arxiv.org/abs/2212.10481>.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. Chain-of-thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems*, 35:24824–24837, 2022.
- Nancy Xu, Sam Masling, Michael Du, Giovanni Campagna, Larry Heck, James Landay, and Monica Lam. Grounding open-domain instructions to automate web support tasks. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pp. 1022–1032, Online, 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.naacl-main.80. URL <https://aclanthology.org/2021.naacl-main.80>.
- Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. HotpotQA: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pp. 2369–2380, Brussels, Belgium, 2018. Association for Computational Linguistics. doi: 10.18653/v1/D18-1259. URL <https://aclanthology.org/D18-1259>.
- Shunyu Yao, Howard Chen, John Yang, and Karthik Narasimhan. Webshop: Towards scalable real-world web interaction with grounded language agents. volume abs/2207.01206, 2022a. URL <https://arxiv.org/abs/2207.01206>.
- Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. React: Synergizing reasoning and acting in language models. *ArXiv preprint*, abs/2210.03629, 2022b. URL <https://arxiv.org/abs/2210.03629>.
- Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas L Griffiths, Yuan Cao, and Karthik Narasimhan. Tree of thoughts: Deliberate problem solving with large language models. *ArXiv preprint*, abs/2305.10601, 2023. URL <https://arxiv.org/abs/2305.10601>.
- Victor Zhong, Caiming Xiong, and Richard Socher. Seq2sql: Generating structured queries from natural language using reinforcement learning. arxiv 2017. *ArXiv preprint*, abs/1709.00103, 2017. URL <https://arxiv.org/abs/1709.00103>.
- Shuyan Zhou, Pengcheng Yin, and Graham Neubig. Hierarchical control of situated agents through natural language. In *Proceedings of the Workshop on Structured and Unstructured Knowledge Integration (SUKI)*, pp. 67–84, Seattle, USA, 2022a. Association for Computational Linguistics. doi: 10.18653/v1/2022.suki-1.8. URL <https://aclanthology.org/2022.suki-1.8>.
- Shuyan Zhou, Li Zhang, Yue Yang, Qing Lyu, Pengcheng Yin, Chris Callison-Burch, and Graham Neubig. Show me more details: Discovering hierarchies of procedures from semi-structured web data. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 2998–3012, Dublin, Ireland, 2022b. Association for Computational Linguistics. doi: 10.18653/v1/2022.acl-long.214. URL <https://aclanthology.org/2022.acl-long.214>.

A APPENDIX

A.1 WEBSITE IMPLEMENTATION

Given the selected websites described in §2.2, we make the best attempt to reproduce the functionality of commonly used sites in a reproducible way. To achieve this, we utilized open-source frameworks for the development of the websites across various categories and imported data from their real-world counterparts. For the E-commerce category, we constructed a shopping website with approximately 90k products, including the prices, options, detailed product descriptions, images, and reviews, spanning over 300 product categories. This website is developed using Adobe Magento, an open-source e-commerce platform⁴. Data resources were obtained from data from actual online sites, such as that included in the Webshop data dump Yao et al. (2022a). As for the social forum platform, we deployed an open-source software Postmill⁵, the open-sourced counterpart of Reddit⁶. We sampled from the top 50 subreddits⁷. We then manually selected many subreddit for northeast US cities as well as subreddit for machine learning and deep learning-related topics. This manual selection encourages cross-website tasks such as seeking information related to the northeast US on both Reddit and the map. In total, we have 95 subreddits, 127390 posts, and 661781 users. For the collaborative software development platform, we choose GitLab⁸. We heuristically simulate the code repository characteristics by sampling at least ten repositories for every programming language: 80% of them are sampled from the set of top 90 percentile wrt stars repos using a discrete probability distribution weighted proportional to their number of stars; the remaining are sampled from the bottom ten percentile set using similar weighted distribution. This is done to ensure fair representation of repos of all kinds, from popular projects with many issues and pull requests to small personal projects. In total, we have 300 repositories and more than 1000 accounts with at least one commit to a repository. For the content management system, we adapted Adobe Magento’s admin portal, deploying the sample data provided in the official guide. We employ OpenStreetMap⁹ for map service implementation, confining our focus to the northeast US region due to data storage constraints. We implement a calculator and a scratchpad ourselves.

Lastly, we configure the knowledge resources as individual websites, complemented with search functionality for efficient information retrieval. Specifically, we utilize Kiwix¹⁰ to host an offline version of English Wikipedia with a knowledge cutoff of May 2023. The user manuals for GitLab and Adobe Commerce Merchant documentation are scraped from the official websites.

A.2 ENVIRONMENT DELIVERY AND RESET

One goal for our evaluation environment is ease of use and reproducibility. As a result, we deploy our websites in separate Docker images¹¹, one per website. The Docker images are fully self-contained with all the code of the website, database, as well as any other software dependencies. They also do not rely on external volume mounts to function, as the data of the websites are also part of the docker image. This way, the image is easy to distribution containing all the pre-populated websites for reproducible evaluation. End users can download our packaged Docker images and run them on their systems and re-deploy the exact websites together with the data used in our benchmarks for their local benchmarking.

Since some evaluation cases may require the agent to modify the data contained in the website, *e.g.*, creating a new user, deleting a post, etc., it is crucial to be able to easily reset the website environment to its initial state. With Docker images, the users could stop and delete the currently running containers for that website and start the container from our original image again to fully reset the environment to the initial state. Depending on the website, this process may take from a few seconds to one minute. However, not all evaluation cases would require an environment reset, as

⁴<https://github.com/magento/magento2>

⁵<https://postmill.xyz/>

⁶<https://www.reddit.com/>

⁷<https://redditlist.com/sfw.html>

⁸<https://gitlab.com/gitlab-org/gitlab>

⁹<https://www.openstreetmap.org/>

¹⁰<https://www.kiwix.org/en/>

¹¹<https://www.docker.com/>

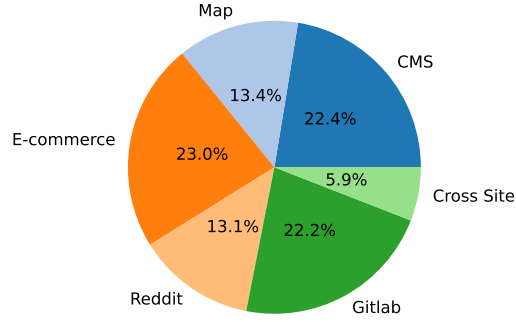


Figure 6: The intent distribution across different websites. Cross-site intents necessitate interacting with multiple websites. Notably, regardless of the website, all user intents require interactions with multiple web pages.

many of the intents are information gathering and are read-only for the website data. Also, combined with the inference time cost for the agent LLMs, we argue that this environment reset method, through restarting Docker containers from the original images, will have a non-negligible but small impact on evaluation time.

A.3 USER ROLES SIMULATION

Users of the same website often have disparate experiences due to their distinct *roles*, *permissions*, and *interaction histories*. For instance, within an E-commerce CMS, a shop owner might possess full read and write permissions across all content, whereas an employee might only be granted write permissions for products but not for customer data. We aim to emulate this scenario by generating unique user profiles on each platform.

On the shopping site, we created a customer profile that has over 35 orders within a span of two years. On GitLab, we selected a user who maintains several popular open-source projects with numerous merge requests and issues. This user also manages a handful of personal projects privately. On Reddit, our chosen profile was a user who actively participates in discussions, with many posts and comments. Lastly, on our E-commerce CMS, we set up a user profile for a shop owner who has full read-and-write access to all system contents.

All users are automatically logged into their accounts using a pre-cached cookie. To our best knowledge, this is the first publicly available agent evaluation environment to implement such a characteristic. Existing literature typically operates under the assumption of universally identical user roles Shi et al. (2017); Liu et al. (2018); Deng et al. (2023).

A.4 INTENT DISTRIBUTION

The distribution of intents across the websites are shown in Figure 6.

A.5 EXPERIMENT CONFIGURATIONS

We experiment with GPT-3.5-TURBO-16K-0613, GPT-4-0613, and TEXT-BISON-001 with a temperature of 1.0 and a top- p parameter of 0.9. The maximum number of state transitions is set to 30. We halt execution if the same action is repeated more than three times on the same observation or if the agent generates three consecutive invalid actions. These situations typically indicate a high likelihood of execution failure and hence warrant early termination. For TEXT-BISON-001, we additionally allow ten retries until it generates a valid action.

Primarily, we use a high temperature of 1.0 to encourage the *exploration*. To aid replicating the results, we provide the results of GPT-3.5-TURBO-16K-0613 with temperature 0.0 in Table 5 and the execution trajectories in our code repository.

CoT	UA	Hint	Model	SR
✓	✗		GPT-3.5	6.28

Table 5: The task success rate (SR %) of GPT-3.5-TURBO-16K-0613 with temperature 0.0.

A.6 PROMPT FOR FUZZY_MATCH

Help a teacher to grade the answer of a student given a question. Keep in mind that the student may use different phrasing or wording to answer the question. The goal is to evaluate whether the answer is semantically equivalent to the reference answer.

question: {{intent}}

reference answer: {{reference answer}}

all the string 'N/A' that you see is a special sequence that means 'not achievable'

student answer: {{prediction}}

Conclude the judgement by correct/incorrect/partially correct.

Predictions that are judged as “correct” will receive a score of one, while all other predictions will receive a score of zero.

A.7 THE PROMPTS OF THE BASELINE WEB AGENTS

The system message of the reasoning agent for both GPT-3.5 and GPT-4 is in Figure 7, and two examples are in Figure 8. The system message of the direct agent for GPT-3.5 is in Figure 9 and the two examples are in Figure 10. **UA hint** refers to the instruction of “If you believe the task is impossible to complete, provide the answer as "N/A" in the bracket.”. We remove this sentence in our ablation studies.

A.8 ADDITIONAL ERROR ANALYSIS

Observation Bias Realistic websites frequently present information on similar topics across various sections to ensure optimal user accessibility. However, a GPT-4 agent often demonstrates a tendency to latch onto the first related piece of information it encounters without sufficiently verifying its relevance or accuracy. For instance, the homepage of the E-Commerce CMS displays the best-selling items based on *recent purchases*, while historical best-seller data is typically accessed via a separate report. Presented with the task of “What is the top-1 best-selling product in 2022”, the GPT-4 agent defaults to leveraging the readily available information on the homepage, bypassing the necessary step of generating the report to obtain the accurate data.

Failures in Observation Interpretation Interestingly, while GPT-4 is capable of summarizing the observations, it occasionally overlooks more granular information, such as the previously entered input. As in the right-hand example of Figure 11, [5172] StaticText indicates that the search term “DMV area” has already been entered. However, the agent disregards this detail and continuously issues the command `type [2430] [DMV area]` until it reaches the maximum step limit. Furthermore, the agent often neglects the previous action information that is provided alongside the observation.

We hypothesize that these observed failures are related to the current pretraining and supervised fine-tuning on dialogues employed in GPT models Ouyang et al. (2022). These models are primarily trained to execute instructions given *immediate* observations (*i.e.*, the dialogue history); thereby, they may exhibit a lack of explorations. Furthermore, in dialogue scenarios, subtle differences in NL expressions often have less impact on the overall conversation. As a result, models may tend to overlook minor variations in their observations.

You are an autonomous intelligent agent tasked with navigating a web browser. You will be given web-based tasks. These tasks will be accomplished through the use of specific actions you can issue.

Here's the information you'll have:

The user's objective: This is the task you're trying to complete.

The current web page's accessibility tree: This is a simplified representation of the webpage, providing key information.

The current web page's URL: This is the page you're currently navigating.

The open tabs: These are the tabs you have open.

The previous action: This is the action you just performed. It may be helpful to track your progress.

The actions you can perform fall into several categories:

Page Operation Actions

`click [id]` : This action clicks on an element with a specific id on the webpage.

`type [id] [content] [press\_enter\_after=0|1]` : Use this to type the content into the field with id. By default, the "Enter" key is pressed after typing unless `press_enter_after` is set to 0.

`hover [id]` : Hover over an element with id.

`press [key\_comb]` : Simulates the pressing of a key combination on the keyboard (e.g., Ctrl+v).

`scroll [direction=down|up]` : Scroll the page up or down.

Tab Management Actions:

`new\_tab` : Open a new, empty browser tab.

`tab\_focus [tab\_index]` : Switch the browser's focus to a specific tab using its index.

`close\_tab` : Close the currently active tab.

URL Navigation Actions:

`goto [url]` : Navigate to a specific URL.

`go\_back` : Navigate to the previously viewed page.

`go\_forward` : Navigate to the next page (if a previous

`go\_back` action was performed).

Completion Action:

`stop [answer]` : Issue this action when you believe the task is complete. If the objective is to find a text-based answer, provide the answer in the bracket. If you believe the task is impossible to complete, provide the answer as "N/A" in the bracket.

Homepage:

If you want to visit other websites, check out the homepage at <http://homepage.com>. It has a list of websites you can visit.

<http://homepage.com/password.html> lists all the account names and passwords for the websites. You can use them to log in to the websites.

To be successful, it is very important to follow the following rules:

1. You should only issue an action that is valid given the current observation
2. You should only issue one action at a time.
3. You should follow the examples to reason step by step and then issue the next action.
4. Generate the action in the correct format. Start with a "In summary, the next action I will perform is" phrase, followed by action inside `````. For example, "In summary, the next action I will perform is ```click [1234]```".
5. Issue stop action when you think you have achieved the objective. Don't generate anything after stop."""

Figure 7: The `system` message of the reasoning agent. This message has a general explanation of the task, the available actions, and some notes on avoiding common failures.

example_user
OBSERVATION:
 [1744] link 'HP CB782A#ABA 640 Inkjet Fax Machine (Renewed)'
 [1749] StaticText '\$279.49'
 [1757] button 'Add to Cart'
 [1760] button 'Add to Wish List'
 [1761] button 'Add to Compare'
 URL: <http://onestopmarket.com/office-products/office-electronics.html>
 OBJECTIVE: What is the price of HP Inkjet Fax Machine
 PREVIOUS ACTION: None

example_assistant
 Let's think step-by-step. This page lists the information of HP Inkjet Fax Machine, which is the product identified in the objective. Its price is \$279.49. I think I have achieved the objective. I will issue the stop action with the answer. In summary, the next action I will perform is `` ` stop [\$279.49]` ``

example_user
OBSERVATION:
 [164] textbox 'Search' focused: True required: False
 [171] button 'Go'
 [174] link 'Find directions between two points'
 [212] heading 'Search Results'
 [216] button 'Close'
 URL: <http://openstreetmap.org>
 OBJECTIVE: Show me the restaurants near ABC
 PREVIOUS ACTION: None

example_assistant
 Let's think step-by-step. This page has a search box whose ID is [164]. According to the nominatim rule of openstreetmap, I can search for the restaurants near a location by 'restaurants near:'. I can submit my typing by pressing the Enter afterwards. In summary, the next action I will perform is `` ` type [164] [restaurants near ABC] [1]` ``

Figure 8: The two examples provided as `example_user` and `example_assistant` for the reasoning agent. Before issuing the action, the agent first perform reasoning.

You are an autonomous intelligent agent tasked with navigating a web browser. You will be given web-based tasks. These tasks will be accomplished through the use of specific actions you can issue.

Here's the information you'll have:
 The user's objective: This is the task you're trying to complete.
 The current web page's accessibility tree: This is a simplified representation of the webpage, providing key information.
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 The previous action: This is the action you just performed. It may be helpful to track your progress.

The actions you can perform fall into several categories:

Page Operation Actions

- `click [id]` : This action clicks on an element with a specific id on the webpage.
- `type [id] [content] [press\_enter\_after=0|1]` : Use this to type the content into the field with id. By default, the "Enter" key is pressed after typing unless `press_enter_after` is set to 0.
- `hover [id]` : Hover over an element with id.
- `press [key\_comb]` : Simulates the pressing of a key combination on the keyboard (e.g., Ctrl+v).
- `scroll [direction=down|up]` : Scroll the page up or down.

Tab Management Actions:

- `new\_tab` : Open a new, empty browser tab.
- `tab\_focus [tab\_index]` : Switch the browser's focus to a specific tab using its index.
- `close\_tab` : Close the currently active tab.

URL Navigation Actions:

- `goto [url]` : Navigate to a specific URL.
- `go\_back` : Navigate to the previously viewed page.
- `go\_forward` : Navigate to the next page (if a previous
- `go\_back` action was performed).

Completion Action:

- `stop [answer]` : Issue this action when you believe the task is complete. If the objective is to find a text-based answer, provide the answer in the bracket. If you believe the task is impossible to complete, provide the answer as "N/A" in the bracket.

Homepage:

If you want to visit other websites, check out the homepage at <http://homepage.com>. It has a list of websites you can visit.
<http://homepage.com/password.html> lists all the account name and password for the websites. You can use them to log in to the websites.

To be successful, it is very important to follow the following rules:
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1. You should only issue an action that is valid given the current observation
2. You should only issue one action at a time.
3. Generate the action in the correct format. Always put the action inside a pair of ```. For example, ```click [1234]```
4. Issue stop action when you think you have achieved the objective. Don't generate anything after stop. ""

Figure 9: The `system` message of the direct agent. This message has the general explanation of the task, the available actions and some notes on avoiding common failures.

example_user
OBSERVATION:
 [1744] link 'HP CB782A#ABA 640 Inkjet Fax Machine (Renewed)'
 [1749] StaticText '\$279.49'
 [1757] button 'Add to Cart'
 [1760] button 'Add to Wish List'
 [1761] button 'Add to Compare'
 URL: http://onestopmarket.com/office-products/office-electronics.html
 OBJECTIVE: What is the price of HP Inkjet Fax Machine
 PREVIOUS ACTION: None

example_assistant
 ``` stop [\$279.49] ```

**example\_user**  
**OBSERVATION:**  
 [164] textbox 'Search' focused: True required: False  
 [171] button 'Go'  
 [174] link 'Find directions between two points'  
 [212] heading 'Search Results'  
 [216] button 'Close'  
 URL: http://openstreetmap.org  
 OBJECTIVE: Show me the restaurants near ABC  
 PREVIOUS ACTION: None

**example\_assistant**  
 ``` type [164] [restaurants near ABC] [1] ```

Figure 10: The two examples provided as `example_user` and `example_assistant` for the direct agent. The agent directly emits the next action given the observation.

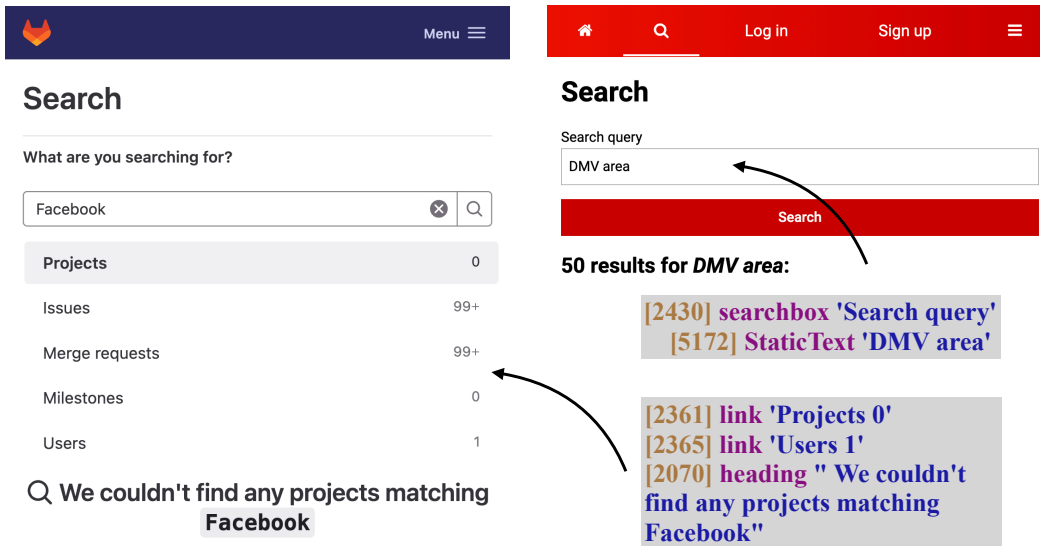


Figure 11: Two examples where the GPT-4 agent failed, along with their screenshot and the accessibility tree of the relevant sections (grey). On the left, the agent fails to proceed to the “Users” section to accomplish the task of “Fork all Facebook repos”; on the right, the agent repeats entering the same search query even though the observation indicates the input box is filled.